CHAPTER ONE

Environmental behavior of glyphosate in soils

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Contents

Abstract

Glyphosate [N-(phosphonomethyl) glycine] (GPS) is currently, and has been for some time, the most commonly applied herbicide worldwide. As such, considerable effort has been put forth to investigate the herbicide's behavior in the environment. In this review, an overview of the existing literature involving GPS is provided, with emphasis placed on recent studies completed in the past 10 years. As this chapter will illustrate, experimental results are highly variable and often contradictory, indicating a need for the behavior of GPS to be considered within the context of a set of specific environmental conditions. The affinity of a soil for solvated GPS is highly dependent upon the physiochemical properties of the matrix, although the herbicide is generally considered as strongly sorbed in most cases. Although the results of laboratory studies suggest that the mobility of GPS is highly limited in soils, the environmental occurrence of the herbicide in surface and groundwater is well documented. Leaching of GPS may be of concern in field soils with well-developed structure, where preferential flow through

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macropores is often responsible for rapid vertical transport. Recent studies indicated that the transport of GPS along with windblown sediment may also be a significant mechanism of off-site deposition. GPS has been traditionally considered as essentially non-toxic; however, more contemporary evidence suggests that it does indeed impart toxic effects to certain aquatic organisms as well as vertebrates. Overall, results presented here suggest that traditional presumptions of the immobility and non-toxicity of GPS may need to be re-evaluated.

1. Introduction and general description

Glyphosate [N-(phosphonomethyl) glycine] (GPS) is a broad-spectrum, non-selective, post emergence herbicide with very high water solubility (12 g/L) (Franz et al., 1997; Maqueda et al., 2017; Sprankle et al., 1975). It works by inhibiting the function of 5-enolpyruvylshikimic acid-3-phosphate synthase, which is an intermediate enzyme involved in the synthesis of aromatic amino acids via the shikimic acid pathway (Boocock and Coggins, 1983). These amino acids serve as essential components in the synthesis of proteins as well as the production of several secondary plant products such as phenolics, lignin, and various growth promoters and inhibitors (Duke and Powles, 2008; Franz et al., 1997). This pathway is non-existent in most living organisms with the exception of plants and certain bacteria and fungi (Bentley and Haslam, 1990). GPS was introduced by the Monsanto Corporation as the active ingredient in the herbicidal formulation Roundup and was made commercially available starting in 1974 (Franz et al., 1997). Since then, its use in agricultural and non-agricultural settings has steadily increased from a total of 0.6 Mg applied in 1974 to a total of 125.5 Mg applied in 2014, and is currently the most widely used herbicide in the United States and throughout the world (Benbrook, 2016; Grandcoin et al., 2017). The introduction of genetically engineered "Roundup Ready" varieties starting in 1996 (Dill, 2005) has certainly contributed to this proliferation of the use of GPS based herbicidal formulations, as weed management via GPS application could be continued throughout the growing season with no detrimental effect on the cultivar.

Glyphosate is a polar, amphoteric molecule characterized by three main functional groups (Borggaard and Gimsing, 2008). These are the phosphonomethyl, amine and carboxymethyl groups arranged in a linear manner. GPS is a polyprotic acid with four pK_a values (p$K_{a1} = 0.8$, p$K_{a2} = 2.23$, p$K_{a3} = 5.46$, p$K_{a4} = 10.14$) (Liu et al., 2016), meaning that speciation of the molecule is dependent upon the pH value of the solution. Over the pH

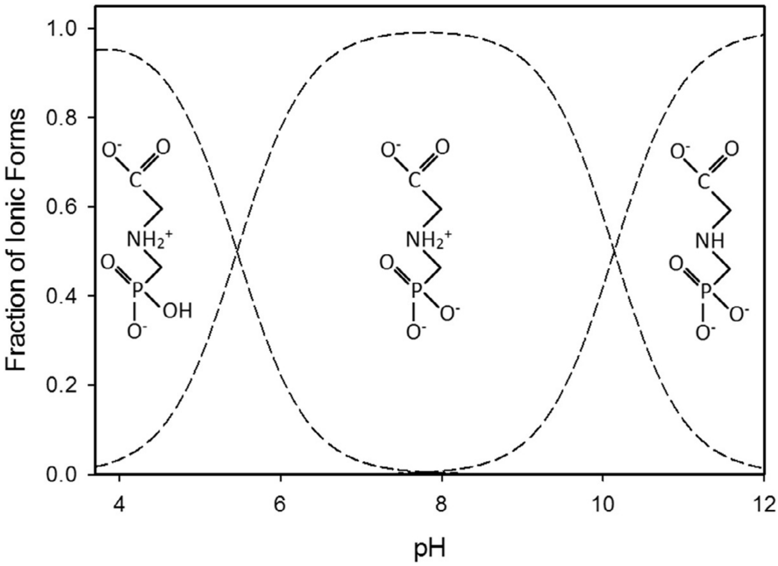

Fig. 1 Fraction of glyphosate species present in solution across a range of pH values. *Redrawn with permission from Borggaard, O.K. and Gimsing, A.L. 2008. Fate of glyphosate in soil and the possibility of leaching to ground and surface waters: a review. Pest. Manag. Sci. 64, 441–456. Copyright 2007, Society of Chemical Industry.*

values commonly found in soils, mono and divalent anions are the predominant species present. Fig. 1 displays the fraction of each species present over a range of typical pH values.

This contribution will review the current state of knowledge on the environmental behavior of GPS, specifically focused on interactions of the herbicide within the soil-water environment. Attention will also be given to reports of the occurrence of GPS in the environment and the modes of transportation that facilitate those observations. Finally, potential toxic effects of the herbicide will be presented, with emphasis placed on soil and aquatic organisms, which are those most frequently exposed to GPS in both timing and magnitude. Evidence of human toxicity will also be acknowledged in order to put presented findings within a larger and more immediate context.

2. Glyphosate in soils

2.1 Retention mechanisms

The equilibrium partitioning of a solvated species into sorbed and solution phases commonly employs the use of various isotherm models. One extensively used model is the Freundlich isotherm, expressed in Eq. (1).

$$S = K_f C^n \tag{1}$$

where S is the sorbed concentration (mg kg^{-1}), K_f is the Freundlich distribution coefficient ($\text{L}^n\ \text{mg}^{1-n}\ \text{kg}^{-1}$), C is the solution concentration (mg L^{-1}), and n is a dimensionless coefficient commonly less than unity. While the majority of GPS applied in agricultural settings ideally remains on plant biomass, a substantial amount is expected to reach the soil where it is generally characterized as a strongly sorbed solute (Borggaard and Gimsing, 2008). Reported Freundlich distribution coefficient values exhibit considerable variation depending upon the soil type, with reported values ranging from 21 to 1892 $\text{L}^n\ \text{mg}^{1-n}\ \text{kg}^{-1}$. Results from a number of recent GPS sorption investigations are briefly summarized in Table 1.

Numerous investigations indicated that sorption onto reactive media is mostly attributed to the phosphonic acid moiety functional group of the GPS molecule (Borggaard and Gimsing, 2008; Glass, 1987; Gros et al., 2017; Sprankle et al., 1975), although evidence exists that interactions involving the carboxylmethyl group may be significant as well (Dideriksen and Stipp, 2003; McBride and Kung, 1989). In soils characterized by physiochemical heterogeneity, amorphous Fe and Al oxides are considered to have the greatest affinity for solvated GPS (Gimsing et al., 2007; Piccolo et al., 1994; Sprankle et al., 1975), where sorption likely occurs mainly via inner-sphere mono and bidentate surface complexation. Conceptualized binding mechanisms of GPS onto Fe oxide (goethite) given by Sheals et al. (2002) are displayed in Fig. 2. Ligand exchange of surface coordinated

Table 1 Recently reported Freundlich isotherm parameters for GPS sorption to soils.

K_f	n	Number of soils used in study	References
80–214	0.56–0.68	23	de Jonge et al. (2001)
21–87	0.69–0.90	6	Albers et al. (2009)
42–78	0.20–0.43	5	Keshteli et al. (2011)
21–149	0.91–1.00	16	Ghafoor et al. (2012)
32–540	0.83–1.09	17	Sidoli et al. (2016)
74–447	0.40–1.21	23	Gros et al. (2017)
81–133	0.81–0.97	4	Maqueda et al. (2017)
37–1892	0.75–0.89	15	Zhelezova et al. (2017)

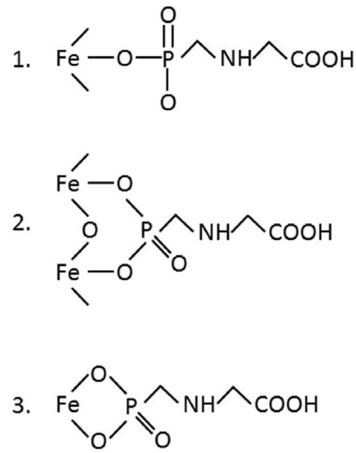

Fig. 2 Proposed binding mechanisms of GPS onto goethite by Sheals et al. (2002): (1) Monodentate mononuclear complex; (2) Bidentate binuclear complex; (3) Bidentate mononuclear complex. *Adapted with permission from Borggaard, O.K. and Gimsing, A.L. 2008. Fate of glyphosate in soil and the possibility of leaching to ground and surface waters: a review. Pest Manag. Sci. 64, 441–456. Copyright 2007, Society of Chemical Industry.*

hydroxyl groups at edge sites of layer silicates also lends to the GPS retentive capacity of soils (Borggaard and Gimsing, 2008; Maqueda et al., 2017), although the contribution of these sites may be less due to a lower overall abundance (Gimsing and Borggaard, 2002). Additionally, GPS sorption has been positively correlated with soil clay content and cation exchange capacity (CEC) (Dollinger et al., 2015; Gros et al., 2017; Paradelo et al., 2015; Sidoli et al., 2016). This positive correlation is likely due to GPS complexation with surface exchangeable polyvalent cations, which would be more prevalent in soils with higher CEC values. As discussed by Dollinger et al. (2015), direct surface association between solvated GPS and negatively charged clay minerals is unlikely due to electrostatic repulsion, however, high retentive capacities of clay rich soils could be attributed to this mechanism. This effect is highly dependent upon the properties of the system. Dousset et al. (2007) reported increased mobility of GPS in granitic soils when Cu was included in the applied solution. The scientists attributed this to Cu-GPS complex formation in solution, which are retained to a lesser extent. If complex formation occurs between the Cu cation and the highly reactive phosphonate group of GPS, this increase in herbicide mobility would be expected. In some instances, GPS may even serve as a mediator for cation sorption to soils. Wang et al. (2009) observed increased Cu^{2+} retention by a Chinese soil when applied in conjunction with GPS.

Assuming the herbicide is bound to the soil matrix via the phosphonate group, it would seem that Cu^{2+} in solution can form stable complexes with the carboxylate group opposite of the binding site. These results emphasize the notion that the full soil solution chemistry must be considered when assessing the mobility potential of GPS in the subsurface environment.

Several studies have shown that GPS sorption onto a variety of materials is a relatively quick process with pseudo-equilibrium conditions being reached in 24 h or less. Gimsing and Borggaard (2002) observed rapid sorption onto goethite and gibbsite, and Piccolo et al. (1994) established a contact time of only 2 h to reach equilibrium conditions in some European soils. Recently, Ozbay et al. (2018) reported equilibrium conditions in <100 min while studying GPS sorption by calcareous soils from Turkey, and Jiang et al. (2018) reported that sorption onto Fe amended biochars was complete in <24 h. However, reports of prolonged kinetics exist as well. Gerritse et al. (1996) reported significant increases in GPS sorption past 24 h in sandy soils from Western Australia, while Gimsing et al. (2004b) observed significant increases in GPS sorption by Danish surface soils over the course of 50 h. A similar result was obtained from sorption studies involving variably charged Tanzanian soil (Gimsing et al., 2007). Furthermore, Padilla and Selim (2019b) observed a time-dependency of GPS sorption by two Louisiana soils for up to 72 h and were able to successfully describe the data with a kinetic model incorporating reversible and irreversible sites. These findings indicate that the effect of kinetics on GPS sorption in soils is highly dependent upon the soil type, and should be considered carefully when assessing the behavior of GPS in the subsurface.

2.2 Role of pH

As GPS is a polyprotic molecule, the behavior of the herbicide in soils is highly dependent upon pH. In general, an inverse relationship exists between the affinity of a soil for solvated GPS and the pH of the soil solution, the reason for which is twofold. As discussed by Gimsing and Borggaard (2007), some variably charged minerals have a high affinity for GPS. As pH increases, these mineral surfaces tend to become more negatively charged and thus, electrostatic attraction to these materials decreases. Similarly, the GPS molecule deprotonates at higher pH and therefore carries a more negative net charge, which enhances electrostatic repulsion between solution phase GPS and permanently charged clay surfaces.

The pH dependence of GPS sorption onto soils has been well documented in the literature since nearly the introduction of the herbicide onto the commercial market (de Jonge and de Jonge, 1999; McConnel and Hossner, 1985; Sprankle et al., 1975). More recently, however, Tèvez and dos Santos Afonso (2015) reported decreased affinities with increasing pH in the three soils studied, which was attributed to changes in the surface charge of the soil particles. Performing sorption studies on Na-montmorillonite, Khoury et al. (2010) found that GPS sorption decreased at both high and low pH values. At pH values <3, both the GPS molecule and variably charged edge sites become positively charged and are therefore electrostatically repelled. The same explanation applies for decreased sorption at high pH values when both the solvated molecule and binding site are negatively charged. Similar findings have been reported by a number of other recent studies (Kumari et al., 2016; Okada et al., 2016; Waiman et al., 2012). While conducting a regression analysis on 17 different agricultural soils in order to predict the sorption potential of GPS, Sidoli et al. (2016) found that pH_{CaCl_2} was the most important explanatory variable to predict Freundlich K_f coefficients. This is consistent with the findings of Paradelo et al. (2015), where clay content and pH were the most common predictors of linear partitioning coefficients in soils sampled across geographical regions. However, when performing similar regression analyses on 101 soil and sediment samples, Dollinger et al. (2015) found that pH was not a significant predictor of linear partitioning or Freundlich K_f coefficients, and was only an explanatory variable for predicting the Freundlich n coefficient (Eq. 1). Although the general trend of decreased GPS sorption by soils with an increase in pH is predominant in the literature, contrasting evidence exists as well. Upon treating soils with lime amendments, de Jonge et al. (2001) reported an increase in GPS sorption. This was attributed to an increase in reactive amorphous Fe/Al oxides with increased pH, which are considered dominant sites for GPS retention. Additionally, Sen et al. (2017) reported that only 42% of solution GPS was removed by a forest soil at pH 2, whereas almost complete removal of GPS was observed at pH 14.

2.3 Influence of soil organic matter

In contrast to several well-studied organic compounds that interact strongly with soil organic matter (SOM), the role of SOM on GPS sorption remains a relatively ambiguous subject. Performing sorption studies on four different purified humic substances, Piccolo et al. (1996) reported a higher affinity of humic substances for solvated GPS than clay minerals. GPS sorption by these materials was positively correlated with the aliphaticity and molecular size of the humic substance. Additionally, Albers et al. (2009) also determined that purified humic substances had a high affinity for GPS; however, higher affinities were positively correlated with the aromatic content of the humic material. Furthermore, it was determined that soil type strongly influenced the role of SOM on GPS sorption, with approximately 40% of total GPS associated with the humic and fulvic acid fractions of a sandy soils, whereas only about 10% of total GPS was associated with the same fractions of a clay soil. When the SOM content was removed from soil samples, Yu and Zhou (2005) found that Freundlich K_f coefficients for bulk soils decreased by as much as 75%. Similar findings were reported by Ololade et al. (2014), who suggested that SOM removal by soil treatment had the potential to remobilize GPS in the subsurface. Dollinger et al. (2015) proposed that reduced affinities of soils for GPS when SOM was removed could be attributed to a decrease in cation exchange capacity, which was found to be a more significant predictor of Freundlich K_f coefficients than organic carbon percentage. Employing quantum chemical modeling techniques, Gros et al. (2017) found that GPS interacts with SOM primarily via H-bonding between organic functional groups and the phosphonic acid moiety, and suggests that GPS in solution is likely in the form of a GPS-SOM complex, a conclusion also reached by others (Daouk et al., 2015). Because H-bonding is generally regarded as a fairly weak interaction, which is unlikely to overcome repulsive electrostatic forces between negatively charged humus and GPS, Borggaard and Gimsing (2008) attribute the GPS—humic substance interaction to the formation of ternary humus-Me-GPS complexes, where "Me" denotes di and trivalent cations in humus samples, the same mechanism postulated by Barrett and McBride (2007).

Conversely, several studies indicated that SOM may inhibit GPS adsorption by soils. Gerritse et al. (1996) found that GPS adsorption was inversely correlated with soil organic carbon, and concluded that SOM does not adsorb the herbicide and that it may in fact compete for adsorption sites on mineral surfaces. Competition between GPS and SOM for active sites was also suggested by Tèvez and dos Santos Afonso (2015), who concluded that functional groups on SOM molecules coordinate with inorganic surfaces in a way similar to that of GPS. The influence of competitive interactions among SOM and GPS was further emphasized by Arroyave et al. (2016) where it was found that the GPS sorption by goethite was almost completely inhibited in the presence of humic acid. This observation was attributed to two mechanisms; humic acid competes for common sites, thereby reducing the total available sites for GPS sorption, and that negatively charged functional groups opposite of the binding site result in enhanced electrostatic repulsion of GPS in solution. This same study also reported increased rates of GPS desorption by humic acid addition, indicating that GPS could potentially be mobilized in a field situation when organic fertilizers are applied or are present.

2.4 Glyphosate and phosphate in soils

Agricultural soils are commonly supplemented with additional phosphorus via fertilizer application, due to the element's function as an essential plant nutrient. As such, it is expected that GPS and inorganic phosphorus (phosphate anion) commonly coexist within these systems. Competitive interactions between the two solutes at the mineral surface have been well recognized for some time (Sprankle et al., 1975). These interactions are attributed to the molecular similarity that exists between the phosphate ion and the highly reactive phosphonomethyl group of the GPS molecule, and the resultant high affinity for the same materials. As the mobility of either solute in the soil environment is of concern, interactions among the two have received substantial attention in the literature.

A number of investigations have demonstrated that soils display a decreased affinity for solvated GPS while in the presence of phosphate. de Jonge et al. (2001) studied GPS sorption under batch conditions by soils that had received historical applications of phosphorus. In soils with the greatest amount of bicarbonate extractable P, Freundlich K_f coefficients decreased some 50% relative to soils with the lowest amounts of extractable P. Here, it was concluded that GPS and phosphorus compete for reactive sites, and that GPS may be more weakly sorbed and therefore, more mobile in soils with high antecedent levels of phosphorus. Munira et al. (2016) also investigated GPS sorption by soils that had received previous applications of inorganic phosphorus fertilizers, and reached similar conclusions. Here, linear partitioning coefficients decreased by 25–44% in soils with the highest Olsen-P concentrations relative to control soils, a result in agreement with those from a more recent study (Munira et al., 2018). While studying the effects of soil redox potential on GPS sorption, Kanissery et al. (2015) also reported a decreased affinity of soils for the herbicide while in the presence of phosphate. Interestingly, the microbially mediated half-life of the herbicide was decreased by the presence of phosphate under anoxic conditions only. A decrease in affinity results in a greater liquid concentrations of GPS, which should facilitate greater rates of microbial degradation due to the enhanced bioavailability of solution-phase herbicide (Eberbach, 1998). The observation of a decreased herbicide half-life under anoxic conditions by phosphate addition suggests that phosphate itself may stimulate GPS degradation in oxygen-limited situations.

Gimsing and Borggaard (2001) investigated competitive interactions between phosphate and GPS at the surface of a pure Fe-oxide (goethite). In their study, it was determined that phosphate in solution can displace previously sorbed GPS, whereas GPS was unable to desorb surface bound phosphate. It was suggested that GPS molecules bound to the surface via the carboxylic acid functional group were more liable to displacement by phosphate, similar to the phosphate facilitated desorption of other surface bound carboxylic acids from goethite (Geelhoed et al., 1998). This same phenomenon was observed in soils, although the effects of competition were less pronounced than in pure mineral systems (Gimsing et al., 2004a). Furthermore, it was apparent that sorption occurred on both competitive and ion specific sites, likely due to a higher degree of mineralogical heterogeneity in these systems which allows for the surface association of GPS via a larger variety of mechanisms. The presence of competitive and ion specific sites was also reported by Gimsing et al. (2007), where competitive surface interactions between phosphate and GPS involving variably charged tropical soils were evaluated. A mechanistic description of these competitive interactions was provided by Gimsing et al. (2004c). In this study, models incorporating time-dependent reversible and irreversible sites, which also accounted for competitive and ion specific adsorption, were able to successfully describe batch data. Additionally, models involving only competitive sites successfully described GPS and phosphate competition in pure mineral systems as well. The mobilization of GPS soil residues by phosphate application can have major implications in agricultural settings. It has been demonstrated that residual GPS mobilized by phosphate fertilizer application can induce phytotoxic effects in soybeans (Bott et al., 2011), as well as in tomatoes (Cornish, 1992). Gomes et al. (2015) contends that this same process has the potential to negatively affect the function of vegetative buffer strips, which serve to minimize the impairment of water bodies adjacent to agricultural operations. The presence of phosphate in the soil solution was reported by Gomes et al. (2016) to enhance root uptake of GPS in willow. However, any phytotoxic effects appeared largely to be negated, likely due to greater antioxidant activity brought about by increased plant phosphorus assimilation.

While a large amount of the work involving GPS/phosphate interactions employs the use of batch methods, studies conducted under flow conditions may provide results more applicable to field conditions. Barrett and McBride (2007) investigated GPS transport through columns with antecedent levels of soil phosphorus, where any effect of competitive interactions was dependent upon the soil type. It was observed that GPS displayed an inability to mobilize phosphate in a course-textured soil; however, phosphate mobility was enhanced by GPS application in dairy and forest soils. Although phosphate mobility was increased in two of the column experiments, it was concluded that only limited competition between the two solutes for reactive sites exists, as phosphate mobility was induced under GPS application greater than recommended agronomic rates. In accordance with the notion that any effect of competition between GPS and phosphate is highly dependent upon soil type, Lu et al. (2005) reported increased GPS mobility by phosphate addition in a red-paddy soil, whereas mobility was decreased in a sandy soil. Phosphate induced effects on herbicide mobility appear to be dependent upon the order of chemical applications to soil as well. In a column transport study involving an agricultural soil from the southeastern United States, phosphate applied in conjunction with and prior to applications of GPS had little effect on the mobility of the herbicide (Padilla and Selim, 2019a). However, phosphate applied subsequent to the application of GPS remobilized a fraction of soil-bound herbicide, resulting in a secondary breakthrough of GPS. These results suggest that GPS residues in soil may be more liable to leaching in field situations where inorganic phosphorus is applied following herbicide applications.

In contrast to reports of lower soil affinities for GPS in the presence of phosphate, Zhao et al. (2009) observed increased retention of GPS by phosphate addition in two of the three soils involved in the study. In this study, soil columns were allowed to drain following an equilibration period, after which, phosphate as KH_2PO_4 was dissolved in the collected drainage. The subsequent solution was then pumped back into the column, which was allowed to equilibrate overnight. This process likely induced a decrease in the pH of the system, which promoted enhanced GPS retention to the extent that effects of competition are likely overcome. Apart from these studies discussed above, investigations involving GPS/phosphate interactions in a transport setting are lacking in the literature.

2.5 Glyphosate and biochar

Biochar has received much attention recently due to several promising benefits associated with its incorporation into agricultural settings. These include increased crop yields (Schmidt et al., 2015), improvements in beneficial properties of soils (Jha et al., 2010), as well as enhanced sequestration of atmospheric carbon (Lehmann et al., 2006) among others. As such, interactions among commonly used agricultural chemicals and biochar are currently the subject of considerable research. The GPS adsorptive behavior of biochar is highly dependent upon the original organic material and pyrolysis temperature. Conducting batch sorption experiments on a variety of biochar materials produced at different pyrolysis temperatures, Hall et al. (2017) determined that GPS sorption by biochars produced at low temperatures (350 °C) was almost negligible for all materials, consistent with the findings of others (Cederlund et al., 2016). However, increasing pyrolysis temperatures resulted in enhanced GPS sorption by all materials despite corresponding pH increases. Additionally, it was determined that pecan and apple hardwood biochar produced at 900 °C had a high affinity for solvated GPS (linear partitioning coefficients of 213 and 216 L kg^{-1}, respectively), whereas cherry hardwood and wood pellet biochars produced at the same temperature displayed lower affinities (linear partitioning coefficients of 22 and 34 L kg^{-1}, respectively). High affinities for hardwood biochar produced at 700–1000 °C were also reported by Mayakaduwa et al. (2016), with optimized values of the Freundlich K_f coefficient of over 7000 L^n mg^{1-n} kg^{-1}. It was therefore suggested that the removal of aqueous phase GPS could be enhanced with biochar. This is a sentiment reiterated by Herath et al. (2016), where it was determined that steam activated rice husk biochar could also effectively remove GPS from solution.

Several studies have indicated that biochar amendment of agricultural soils has the potential to increase GPS retention capacities. Kumari et al. (2016) reported enhanced sorption of GPS by soils amended with biochar, although it was determined that the extent of increased sorption was highly dependent upon the original soil properties. Conducting a greenhouse pot study, Hagner et al. (2013) reported reduced GPS leaching from soils amended with birch wood biochar. Building upon these results, Hagner et al. (2015) found that the reduced leaching of both GPS and AMPA by biochar amendment was correlated with the pyrolysis temperature of the biochar, with enhanced retention by soils amended with biochars produced at higher pyrolysis temperatures. Interestingly, phosphate leaching was not affected by biochar amendment, suggesting that chemical interactions between the herbicide and biochar may not occur via binding with the highly reactive phosphonomethyl group on the GPS and AMPA molecules. This is in agreement with the conclusion reached by Mayakaduwa et al. (2016), where it was suggested that the protonated amine group may be the most reactive constituent of GPS when binding to hardwood biochar materials. However, it was determined that Fe amendments to biochar did increase retentive capacities (Cederlund et al., 2016; Jiang et al., 2018), most likely due to the formation of Fe oxide coatings which have been shown to interact strongly with GPS. This indicates that interactions involving the phosphonomethyl and carboxymethyl groups may still be significant when certain amendments are incorporated. In general, studies involving GPS behavior in biochar-amended soils are lacking and additional work is needed, especially with the expected future increase of biochar use within agricultural settings.

3. Glyphosate in the environment

3.1 Occurrence

Although GPS is considered as a strongly sorbed solute in soils and is therefore regarded as relatively immobile, instances of its occurrence in the environment are well reported in the literature. Considerable attention has been given to the Pampas region of Argentina. While investigating the occurrence of GPS in surface waters in agricultural basins within this area, Berman et al. (2018) detected the herbicide at >40% frequency in samples of lake water, suspended sediment and benthic sediment. Additionally, Primost et al. (2017) reported that GPS was detected at 22–55% frequency in water samples collected from first and second order streams in agricultural watersheds within the same Pampas region, corroborating results reported previously (Lupi et al., 2015). Furthermore, Okada et al. (2018) detected GPS in groundwater samples at 24% frequency. Interestingly, there was no correlation between measured concentrations and water table depth (water table depth ranging from <4 to >8 m), indicating that GPS contamination of relatively shallow and deep groundwater may be of equal concern.

Reports of GPS in the environment from other parts of the world are numerous as well. Van Stempvoort et al. (2014) detected GPS and its main degradation product in the riparian groundwater of 4 out of 5 stream sites in Canada. The reaches that were sampled were all characterized as gaining streams at the time, indicating that the transport of GPS from groundwater to surface waters is significant and of concern. Similar results were obtained from a later study, where the herbicide was detected at approximately 10% frequency in samples from riparian, wetland, and upland groundwater within a rural watershed in Canada (Van Stempvoort et al., 2016).

Conducting a survey of water bodies from across the United States, GPS was detected at approximately 53% frequency in streams and rivers, 34% frequency in lakes, ponds, and wetlands, and 6% frequency in groundwater (Battaglin et al., 2014). Moreover, GPS has been detected in surface and groundwater in agricultural regions of Mexico (Ruiz-Toledo et al., 2014), Portugal (Abrantes et al., 2010), as well as in groundwater samples from Spain (Sanchís et al., 2012). GPS contamination of ground and surface waters is not limited to agricultural use. Bonansea et al. (2018) studied the occurrence of GPS in surface waters in and around Cordoba City, Argentina, and found the greatest concentrations of the herbicide adjacent to a "green belt" within the city. This suggests that herbicide loading into aquatic systems via landscape applications is likely significant. Moreover, GPS was also detected in 8 out of 13 urban sites in Auckland, New Zealand (Stewart et al., 2014), and was found at a 93% frequency in samples collected from French urban storm water outlets (Zgheib et al., 2012).

3.2 Transport in soils

Determining the mobility of solutes in the subsurface is fundamental to assessing their potential environmental hazard. Laboratory miscible displacement studies coupled with numerical modeling techniques can provide invaluable insight upon the physiochemical mechanisms that dictate the behavior of a chemical within a reactive porous matrix under flow conditions. As GPS is one of the most widely applied agro-chemicals worldwide, several investigators have applied this approach to study the herbicide's mobility within agricultural soils. Incorporating variable flow rates ranging from 0.3 to 30 m day^{-1}, Beltran et al. (1998) evaluated the transport of GPS through four sandy soils from Australia and observed enhanced mobility with increasing flow rates, indicating a time-dependency of GPS retention reactions. A successful description of the GPS breakthrough was obtained with the use of transport models incorporating a time-dependent Freundlich K_f coefficient. Candela et al. (2007) conducted a similar study with Spanish soils, where both flow velocities and column length were varied. Similar to the previously described study, the mobility of GPS was highly dependent upon the flow velocity. Only 15% of applied GPS was recovered from 2 cm columns at a flow velocity of 1.25 cm h^{-1}, compared to 90% recovery of the applied herbicide when flow velocities were increased by two orders of magnitude. The effect of column length was also dependent upon flow velocities. There was a significantly lower peak effluent concentration observed from a 7 cm column relative to a 2 cm column with a constant flow rate of 12.5 cm h^{-1}, however, breakthrough curves were nearly identical from columns of different lengths when the flow velocity was increased to 125 cm h^{-1}. In the case of the higher flow velocity (125 cm h^{-1}), interactions between solvated GPS and the soil matrix most likely take place mainly via equilibrium reactions, whereas the effects of chemical kinetics are more pronounced at lower flow velocities. Furthermore, a two-site model incorporating equilibrium and kinetic sites along with a first-order irreversible sink successfully described breakthrough data. A two-site model was also employed to describe GPS breakthrough from two cultivated soils in Louisiana, although only kinetic type reactions were considered (Padilla and Selim, 2018). Here, retention was described using first-order irreversible and fractional order (Freundlich-type) reversible type reactions, and provided a superior description of the data to that obtained with linear modeling. It was subsequently determined that herbicide retention in these soils was dominated by irreversible reactions. Additionally, Magga et al. (2008) applied a two-site model that considered equilibrium and kinetic reactions to describe GPS breakthrough from an 80 cm column applied as a continuous pulse. In this case, however, any irreversible reaction in the system was considered as mass loss due to biodegradation of the herbicide, which was accounted for with additional terms to approximate microbial biomass dynamics based on Haldane kinetics. Moreover, Zhou et al. (2010) applied a two-site model to describe GPS transport through an undisturbed sandy soil from central China. Although the model successfully described the observed data, it is difficult to assess the overall model performance as the dataset is relatively small (6 points), and GPS was not detected in the effluent until the last sample.

Results from other laboratory studies indicated that GPS retention in soils may not be limited only by chemical kinetics, but by the availability of reactive sites as well. Upon an initial application of GPS, Barrett and McBride (2007) observed 85–95% retention of the herbicide in course-textured soil columns. When a subsequent pulse of GPS was applied to the same columns, retention of the herbicide was decreased to 63–73% of the applied mass. It would seem that in this case, a lower proportion of the cumulative amount of reactive sites is available for GPS sorption when the second pulse is applied, resulting in enhanced recovery in the effluent solution. Zhao et al. (2009) investigated the effect of loading rate on the mobility of GPS in soils from the North China Plain. Three influent concentrations corresponding to 30.8, 44.0 and 220.0 mg/column were applied to columns at a constant flow rate. Pronounced retardation of the subsequent GPS breakthrough was observed for lowest input loads (~14–20 pore volumes before peak arrival), whereas relatively rapid breakthrough was associated with the highest input load (~3–5 pore volumes before peak arrival). This result is likely attributed to either GPS association with a higher proportion of lower affinity sites with increasing input concentrations, or that the availability of reactive sites is limited given sufficient herbicide loading.

Although laboratory column experiments are useful for determining the physiochemical mechanisms that govern solute transport in soils, they are conducted under highly idealized conditions and therefore, may not necessarily be representative of transport in a field setting. As such, several studies have been conducted in order to determine the mobility of GPS under field conditions. Kjær et al. (2011) monitored the leachate from 1 m below tile-drained Danish soils over the course of an 8-month field study. GPS was detected in concentrations exceeding those mandated by EU water quality standards for up to 7 days following precipitation events, which was attributed primarily to preferential flow in macropores. Rapid transport of GPS in macropores was also reported by de Jonge et al. (2000) while studying the leaching potential of undisturbed columns, as well as in an earlier field study (Kjær et al., 2005). The findings of these studies are significant, as they indicate that GPS mobility may be of greater concern in soils with well-developed structure, particularly in areas with relatively shallow water tables.

Despite evidence of GPS mobility under field conditions, a number of investigations have indicated highly limited mobility of the herbicide as well. Al-Rajab et al. (2008) conducted a lysimeter study where leachates collected at 25 cm depth were monitored over an 11-month period. Of the applied herbicide mass, only 0.28% was recovered, resulting in the author's conclusion that ground-water contamination by GPS was not a concern. Extremely limited recovery in leachate solutions collected from field sites was also reported by Bergström et al. (2011), where <0.02% of the applied herbicide mass was recovered from 1.05 m below the ground surface over a period of 748 days. Surprisingly, nearly 60% of the initial GPS was recovered in soil residues as either GPS or AMPA after 748 days, emphasizing the fact that while high soil affinities may render the herbicide as practically immobile, persistence of the chemical in the subsurface environment may be prolonged to a high extent. Consistent with the above results, Siimes et al. (2006) observed negligible GPS in subsurface leachates collected at 1 m depths over 302 days, and Okada et al. (2016) recovered only 0.24% of applied GPS in the effluent from undisturbed soil columns of 15 cm in length.

While it is generally considered that the liquid and gas phases are the only mobile components of the soil system, it may be necessary in certain cases to consider some fraction of the soil matrix as mobile as well (de Jonge et al., 2004). Especially in the case of strongly sorbing solutes, which are expected to be relatively immobile in soils, a significant amount of off-site transference may be attributed to colloid-facilitated transport. Several studies have demonstrated that this is indeed the case with GPS. In a flume study investigating the contribution of erosive processes to GPS transport, Yang et al. (2015) found that 14% of the applied herbicide was transferred off-site with the surface runoff, with the vast majority associated with the suspended load. Furthermore, 72% of the applied GPS was recovered from the uppermost 2 cm of the soil profile (total depth of 35 cm), indicating that lateral surface transport was more significant than vertical transport. Bento et al. (2018) also conducted a flume experiment to investigate the effects of small-scale surface topography on GPS transport as related to soil erosion. Nearly 18% of the applied herbicide was transported off-site with the suspended load with a smooth soil surface, whereas colloid-facilitated transport only accounted for the off-site mobility of 9.4% of the applied GPS when "seeding lines" were included on the soil surface. These two studies demonstrate that particle-facilitated transport of GPS may be significant and of concern in areas that are susceptible to rainfall erosion processes. Vertical transport of soil bound residues through macropores was also cited as a mechanism for the observed GPS concentrations in leachates collected from field studies (Kjær et al., 2011) and undisturbed column studies (de Jonge et al., 2000), described above.

Another mechanism of particle-facilitated transport of GPS that has received little attention until recently is transference of the herbicide along with wind-blown sediment. The first systematic atmospheric measurements of GPS and AMPA in the United States were obtained by Chang et al. (2011), conducted in agricultural regions of Mississippi, Iowa, and Indiana. In this study, GPS was detected at a frequency of >60% in air and rain samples from all three sites, and AMPA was detected at a frequency of >50%. Because of the non-volatile nature of both compounds, introduction of the chemicals to the atmosphere can only occur via spray drift or by wind-blown sediment. As AMPA is a degradation product of GPS that can only be formed in the soil, any atmospheric detection of this compound is attributed specifically to wind-blown sediment. Similar results obtained from an agricultural field in Malaysia were reported the same year by Morshed et al. (2011). In this case, air samples from 1 m above the ground surface at the edges of the test plot were taken using a variety of measurement techniques prior and subsequent to the application of GPS. Since GPS was only detected in quartz filter samples, herbicide residues present in the air column were associated with solid particles and were not in vapor form. More recently, Aparicio et al. (2018) detected GPS in wind-blown sediment collected at 13.5, 50, and 150 cm above an agricultural field in Argentina. Interestingly, the material collected at 150 cm had concentrations of GPS that were 60 times those measured in the bulk soils at the ground surface, indicating that the herbicide residues are concentrated in the very fine material that is more liable to mobilization by wind. Enrichment of GPS in the finer textured fraction of soils was also reported by Mendez et al. (2017) and Bento et al. (2017), resulting in both studies concluding that there is a high risk of off-site transport by wind-blown material. It is therefore proposed here that the widespread environmental occurrence of GPS may largely be attributed to transference of the herbicide off-site by eolian processes. However, a literature search reveals that research in this area is lacking, and further work is required to determine the magnitude of this type of contribution to environmental concentrations of the herbicide.

3.3 Dissipation

Several studies have indicated negligible GPS dissipation under sterile conditions (Bento et al., 2016; Kanissery et al., 2015; Rueppel et al., 1977) leading to the conclusion that the environmental degradation of the herbicide is a microbial process (Sviridov et al., 2015). GPS undergoes biological degradation via two main metabolic pathways resulting in the production of the primary metabolites aminomethylphosphonic acid (AMPA) and sarcosine. Within the AMPA degradation pathway, the first step involves the cleaving of the carboxyl group from the initial GPS molecule via the glyphosate oxidoreductase enzyme, resulting in production of AMPA and glyoxylate. AMPA is subsequently degraded to inorganic phosphate and methylamine, whereas glyoxylate is ultimately degraded within the glyoxylate cycle. The sarcosine pathway is initiated by the cleaving of the C—P bond by the C—P lyase enzyme, resulting in the production of inorganic phosphate and sarcosine. Sarcosine is further degraded to glycine by the sarcosine oxidase enzyme, where the final environmental fate of glycine is complete mineralization to ammonium and carbon dioxide. A schematic representation of the microbial degradation of GPS can be seen in Fig. 3.

AMPA is ubiquitous in soils to which GPS has been applied whereas sarcosine has not been detected (Borggaard and Gimsing, 2008). This is

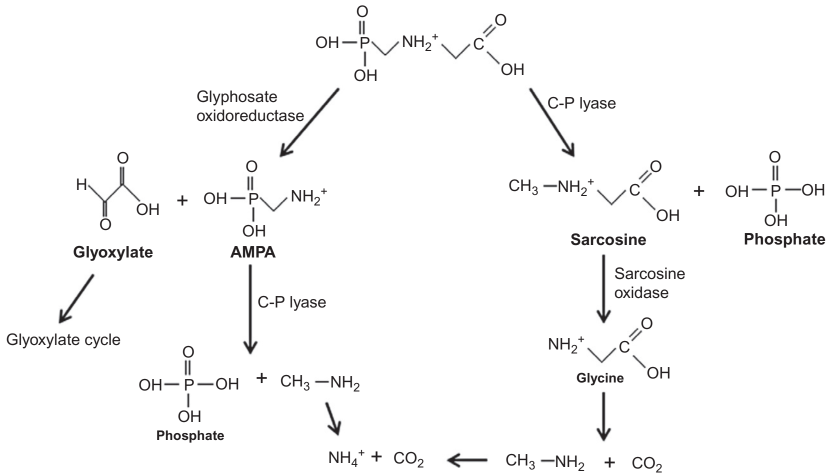

Fig. 3 Microbial degradation of GPS via the AMPA and Sarcosine pathways. *Adapted from Grandcoin, A., Piel, S. and Baures, E., 2017. AminoMethylPhosphonic acid (AMPA) in natural water: its sources, behavior, and environmental fate, Water Res. 117, 187–197, with permission from Elsevier.*

despite the fact that bacteria capable of degrading GPS through the sarcosine pathway have been isolated (Kishore and Jacob, 1987; Sviridov et al., 2015). The absence of sarcosine in soils may be attributed to its rapid degradation or leaching through the soil profile, as the lack of the highly reactive phosphonate group would promote the mobility of this compound. Additionally, the activity of the C—P lyase enzyme is induced under conditions where exogenous phosphorus is low (Sviridov et al., 2015), meaning that the sarcosine degradation pathway could be dominant in soils in which P is a limiting nutrient.

The rate and extent of GPS mineralization is highly variable and is dependent upon several soil factors, with reported half-life values ranging from 4 to 835 days (Table 2). von Wirèn-Lehr et al. (1997) correlated GPS degradation in agricultural soils with total microbial biomass, a result also reported by other publications (Franz et al., 1997; Rueppel et al., 1977). Conversely, Gimsing et al. (2004b) found no correlation between overall microbial biomass and GPS degradation, but instead related degradation with the abundance of *Pseudomonas* spp. bacterial populations. The affinity of a soil for GPS also plays an important role in the degradation of the herbicide, as a high affinity may limit its bioavailability. Al-Rajab and Schiavon (2010) found an inverse relationship between degradation rates and the sorption capacity of soils, indicating that solution phase GPS

Table 2 Reported GPS half-life values in soil.

Half-life (days)	References
6–9 (Labile)	Eberbach (1998)
222–835 (Non-Labile)	
9	Simonsen et al. (2008)
4–19	Al-Rajab and Schiavon (2010)
31	Al-Rajab and Hakami (2014)
15–18 (Oxic Conditions)	Kanissery et al. (2015)
42–51 (Anoxic Conditions)	
17–87	Zhelezova et al. (2017)

is more liable to microbial degradation than sorbed phase GPS. Nguyen et al. (2018b) correlated GPS mineralization in 21 different soils with varying properties and determined that there was a highly significant negative correlation between cumulative GPS mineralization and NaOH extractable residues. These residues are taken to be representative of the strongly sorbed, and therefore less bioavailable, herbicide pool. Additionally, Zhelezova et al. (2017) reported a positive correlation between Freundlich distribution coefficients and GPS half-life values. The redox potential of a soil also has significant effects on GPS mineralization. Rueppel et al. (1977) reported reduced recovery of $^{14}CO_2$ from soils incubated under anaerobic conditions relative to aerobically treated soils, although degradation under both conditions was significant and rapid. Kanissery et al. (2015) also studied the effect of soil redox conditions on GPS degradation, and found that half-life values were increased within all three soils involved in the study under anoxic conditions. The implications of this finding are significant, as GPS may be more persistent in environments experiencing periods of relatively long-term inundation.

The kinetics of GPS degradation is generally characterized by an initial rapid phase followed by much slower dissipation. As first-order models adequately describe GPS degradation only in limited cases (Borggaard and Gimsing, 2008), Eberbach (1998) applied two separate first-order models in order to partition GPS into labile and non-labile pools. The initial rapid degradation of the labile pool was attributed to solution phase and readily exchangeable GPS, whereas the slow degradation of the non-labile pool was associated with the specifically sorbed phase. This again emphasizes the importance of soil sorption characteristics on the dissipation/persistence of GPS. Nevertheless, the lack of an initial lag phase has also been reported by several other scientists (Al-Rajab and Hakami, 2014; Forlani et al., 1999; Kanissery et al., 2015). Their results indicate that the metabolic pathways necessary for degradation are already present within the soil micro-fauna and that GPS degradation is likely a co-metabolic process (Borggaard and Gimsing, 2008; Sprankle et al., 1975).

4. Glyphosate toxicity

The US EPA mandated maximum admissible level of glyphosate in drinking water is 700 μg/L (US EPA, 2016), whereas those required by the European Union are not to exceed 0.1 μg/L for any single pesticide and 0.5 μg/L for total pesticides (Grandcoin et al., 2017). Differences between the two water quality standards are due to differing policies, with the European Union employing a set limit for all pesticides and the EPA setting limits specific to GPS based on the results of toxicity studies. However, the toxicity of GPS is a highly contentious issue and no decisive consensus within the literature has been reached.

4.1 Soil microorganisms

As soil microorganisms are an important metric in assessing overall "soil health," the non-target effects of GPS application on micro-flora is an area of substantial interest. Several studies indicated minimal effects of GPS on soil microbial communities. Weaver et al. (2007) reported no significant effect on rhizosphere or bulk soil microbial populations in Mississippi soils following two post emergent GPS applications, and Dennis et al. (2018) found no difference in the microbial functionality or substrate utilization between control soils and those receiving GPS at recommended field application rates. Furthermore, Nguyen et al. (2018a) observed minimal effects on soil respiration and enzyme activities when GPS was applied at recommended rates, however, more apparent effects were induced under greater application rates.

GPS application has been shown to stimulate microbial activity in a number of studies. Haney et al. (2000) observed increased rates of C and N mineralization in soils that received GPS. Here, there was a strong correlation between the amounts of mineralized C and N and the amounts of C and N added as herbicide, suggesting that increased mineralization is directly a result of applied GPS. Additionally, Mijangos et al. (2009) found enhanced activity and functional diversity within the heterotrophic soil microbial community 15 days following GPS application as the commercial formulation Roundup Plus, however, no consistent difference between treated and control soil occurred after 30 days.]Since GPS is degraded relatively quickly in soils, it is likely that its function as a C, N, or P substrate is transient over longer time periods. Lancaster et al. (2010) conducted a study investigating the effect of multiple herbicide applications on GPS mineralization. They observed that although mineralization rates decreased in soils receiving 4 and 5 applications, incorporation of ^{14}C into the microbial biomass increased. This suggests a shift in the composition of the microbial community in soils receiving high inputs of GPS toward a predominance of species that can utilize the herbicide as a C substrate for anabolic activities. These results may explain the findings of enhanced microbial activities upon GPS application in soils with historic exposure to the herbicide reported by other studies (Lane et al., 2012a; Partoazar et al., 2011). Nguyen et al. (2016) provide a summary of the trend of enhanced soil microbial activity upon GPS application; while conducting a meta-analysis of the results obtained from 36 separate studies, the investigators concluded that the response to herbicide application is highly dependent upon soil properties and that any effects are transitory, typically lasting <60 days.

Contrary to the results given above, evidence of detrimental effects of GPS on soil micro-biota exists as well. Lane et al. (2012b) reported decreased microbial biomass in the rhizosphere of soybeans 7 days following GPS application. Zobiole et al. (2011) observed reductions of *Pseudomonas* spp., Mn reducing bacteria, and indole acetic acid (IAA) producing bacteria in rhizosphere soil subsequent to GPS applications. Since IAA production is one of the major plant growth facilitating functions of rhizosphere bacteria (Mohite, 2013), the reported reductions of glyphosate resistant soybean root and shoot biomass in these same soils is likely attributed to decreases in the IAA producing bacterial populations. A number of recent studies have suggested that soil fungi are especially susceptible to GPS application. Druille et al. (2017) reported a 52% decrease in the number of viable spores of arbuscular mycorrhizal fungi in soils receiving GPS relative to control soils. Colonization of the cortical root cells of *Lotus tenuis* by these same fungi was also decreased by 40%. Based on these results, the investigators concluded that herbicide application may not only effect the production of *L. tenuis*, but the overall productivity of the complete plant community as well. Furthermore, Liu et al. (2018) reported that cultivable fungi was decreased by 63% when GPS was applied at $10\times$ field recommended rates, and fungal biomass was decreased by approximately 30% when the herbicide was applied at field recommended rates. Consistent with previously discussed results of enhanced activity with GPS application, an increase in catabolic activity of Gram-negative bacteria was observed at the 10 × recommended rate treatment, however, biomass carbon was reduced by 45% and the number of cultivable bacteria decreased by 84%. A possible explanation for this is enhanced mortality of bacterial cells with low herbicide tolerance upon GPS application, whereupon these dead cells are utilized as a carbon substrate by species with greater resilience, resulting in enhanced catabolism coupled with the lower bacterial biomass and cultivable numbers. In general, results of studies examining the effect of GPS on the soil micro-flora are variable and often contradictory, suggesting the necessity of further investigations.

4.2 Animals

The prolific use of GPS based herbicide formulations in agricultural and non-agricultural settings has raised concerns about possible toxic effects to multi-cellular organisms; especially to those aquatic organisms residing in water bodies that commonly receive agricultural or urban runoff. Milan et al. (2018) exposed a species of mussel to GPS concentrations of 0.01–1 mg L^{-1}, and observed the physiological response over the course of 21 days. It was determined that herbicide induced differential gene expression negatively affected several key biological processes such as energy metabolism and cell signaling, among others. Avigliano et al. (2018) also demonstrated negative effects of GPS at environmentally relevant concentrations. In their study, a species of estuarine crab was exposed to GPS concentrations ranging from 0.02 to 1 mg L^{-1} during a pre-reproductive timeframe, whereupon both somatic and ovarian growth was impaired. Furthermore, Avigliano et al. (2014) reported up to 33% mortality of a crawfish species exposed to 40 mg L^{-1} GPS, although it is uncertain if this high of a concentration is representative of expected environmental levels of the herbicide. Several other examples of GPS toxicity to both aquatic and terrestrial organisms are also well documented (Gill et al., 2018).

Despite the above studies, the toxic effects attributable to GPS seem to be species dependent. Perkins et al. (2000) found that the LC_{50} (liquid concentration inducing 50% mortality) for frog embryos was 5407 mg AE L^{-1} (acid equivalent L^{-1}) GPS, but the worst-case scenario for environmental concentrations of GPS would only be expected to be 2.8 mg AE L^{-1}. This would suggest that GPS is practically non-toxic at expected environmental concentrations. Several other studies indicated that GPS itself may be non-toxic to various aquatic organisms, but toxic effects of commercial formulations can be mainly attributed to surfactants included in the bulk herbicidal mix, specifically polyethoxylated tallowamine (POEA). While comparing the toxicity of POEA and five different GPS formulations on North American frog species, Howe et al. (2009) determined POEA to be the most harmful compound. Based on this, these scientists concluded that the formulation components should be considered when determining the overall toxicity of GPS based herbicides. This general sentiment was reiterated by the findings of Cuhra et al. (2013) and Szèkács et al. (2014), where chronic exposure to Roundup was more toxic to *Daphnia magna* than was exposure to the isopropylamine salt of GPS itself. Frontera et al. (2011) investigated the toxic effects of GPS and POEA, both as individual compounds and in a 3:1 mixture, and found that the mixture induced the greatest harm to juvenile crawfish. This is a significant result, as it suggests a synergistic interaction between the two chemicals that results in a more toxic effect when they are applied concurrently. Moreover, it suggests that the cumulative toxicity of a commercial formulation may be greater than the "sum" of its individual parts. Therefore, it may be more productive to quantify the impairment of a water body due to GPS contamination based on the presence of GPS along with its associated adjuvants.

Human exposure to GPS is likely a more immediate concern to those individuals who directly operate within agricultural settings. While the risks of exposure to GPS herbicides (and any other agro-chemical) during application are well known, and are therefore usually properly addressed, other less obvious modes of exposure exist as well. It has been documented that GPS residues in soils are concentrated in the respirable dust fraction (Mendez et al., 2017) which is easily mobilized into the air column. Thus, the perturbation of the soil by wind or the movement of machinery could be a potential mechanism by which humans can ingest GPS via inhalation. The general public may also be exposed to the herbicide through the consumption of produce with persistent GPS residues. GPS residues were detected at >90% frequency in 300 soybean samples in 2011 (Myers et al., 2016), and may be more prevalent in GPS-resistant cultivars (Bøhn et al., 2014). GPS has even been detected in processed food products such as honey and soy sauce (Rubio et al., 2014), as well as in baked bread (Myers et al., 2016). Krüger et al. (2014) detected GPS in the urine of rabbits, cows, and humans. This same study also detected GPS in the intestinal, liver, muscle, spleen and kidney tissue of slaughtered cows, suggesting that some amount of ingested

GPS is incorporated into animal biomass. This is a considerable finding, as it not only indicates that GPS is likely accumulated into the biological tissues of humans, but that consumption of meat products may also be a significant vector of human exposure to the herbicide.

Although GPS has traditionally been considered as relatively non-toxic to humans, the results of recent toxicity studies may contradict this presumption. Gehin et al. (2005) determined that the GPS based formulation Roundup 3 plus® was responsible for oxidative damage to human epidermal cells, and recommended the addition of various antioxidants to herbicidal mixes to minimize potential hazards. Richard et al. (2005) reported toxic effect of GPS itself at concentrations lower than expected environmental levels on human placental cells after 18h of exposure. This study also found that GPS formulations were more toxic than GPS alone, suggesting that adjuvants included in the bulk mix either are themselves toxic or enhance the toxicity of GPS via a synergistic mechanism. A number of additional studies have also indicated cytotoxic effects of GPS and its associated commercial formulations (Benachour and Seralini, 2009; Koller et al., 2012; Mesnage et al., 2013). Besides cytotoxicity associated with GPS exposure, potential carcinogenic effects have also been documented. Potti and Sehgal (2005) found that exposure to GPS herbicide formulations affected gene expression in human prostate cells, indicating a potential for the growth of cancer cells. Mañas et al. (2009) documented a significant increase in genotoxic effects induced by the exposure of Hep-2 cells to various concentrations of GPS, signifying a possibility for cell mutation during mitosis. Furthermore, the World Health Organization stated in 2015 that GPS is "probably carcinogenic to humans" (Grandcoin et al., 2017). A study performed by Thongprakaisang et al. (2013) determined that GPS is a potential endocrine disruptor in humans. Specifically, it has the potential to inhibit the function of estrogen receptors, resulting in the propagation of hormone-dependent human breast cancer cells. While this study is often cited as evidence for the GPS induction of cancer in humans, it is important to note that the herbicide promoted the growth of already cancerous cells. Due to the potential toxicity associated with GPS, it is imperative to quantify its behavior in the environment such that leaching to ground or surface waters is minimized.

5. Future outlook

The use of GPS based herbicidal formulations along with "Roundup Ready" cultivars has played an important role in commercial agricultural production in the recent past, and will almost certainly continue in this function into the near future. While GPS has a well demonstrated high affinity for most soils, it is critical to acknowledge the widespread occurrence of the herbicide off-site of its initial application area, indicating at least some degree of mobility in the environment. Some portion of these observations is undoubtedly attributed to preferential flow through macropores, as well to colloid-facilitated transport. Recently, however, evidence that GPS residues are concentrated in the finer textured fraction of the bulk soil suggest that transport via eolian processes may be a significant contribution to receiving water bodies, as well as a potential additional vector of human exposure. This mode of transport has received little attention in the literature to date, and should be investigated further. Furthermore, as soil amendments, such as various biochars, are increasingly incorporated into agricultural systems, it is critical to understand their interactions among commonly used agrochemicals. While GPS/biochar interactions have recently been given consideration, this is an area in which further study is needed. Summarily, while GPS is a well-studied compound, it is meaningful to continue to study its behavior in soils such that efforts to predict its environmental fate can be refined and ultimately improved.

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